

Q1 (4)

$$\lim_{x \rightarrow 0} \frac{ae^x - b \cos x + ce^{-x}}{x \sin x} = 2$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{a\left(1+x+\frac{x^2}{2!}+\dots\right) - b\left(1-\frac{x^2}{2!}+\dots\right) + c\left(1-x+\frac{x^2}{2!}\right)}{\left(\frac{x \sin x}{x}\right)x} = 2$$

$$a - b + c = 0$$

$$a - c = 0$$

$$\& \frac{a+b+c}{2} = 2$$

$$\Rightarrow a + b + c = 4$$

Q2 (4)

$$\lim_{x \rightarrow 0^+} \frac{\cos^{-1} x}{(1-x^2)} \times \frac{\sin^{-1} x}{x} = \frac{\pi}{2}$$

Q3 (1)

We know that

$$r \leq [r] < r + 1$$

and

$$2r \leq [2r] < 2r + 1$$

$$3r \leq [3r] < 3r + 1$$

Now,

$$\lim_{n \rightarrow \infty} \frac{n(n+1) \cdot r}{2 \cdot n^2} = \frac{r}{2}$$

and

$$\lim_{n \rightarrow \infty} \frac{\frac{n(n+1)r}{2} + n}{n^2} = \frac{r}{2}$$

So, by Sandwich Theorem, we can conclude that

$$\lim_{n \rightarrow \infty} \frac{[r] + [2r] + \dots + [nr]}{n^2} = \frac{r}{2}$$

Q4 (1)

$$\begin{aligned} & \lim_{\theta \rightarrow 0} \frac{\tan(\pi(1 - \sin^2 \theta))}{\sin(2\pi \sin^2 \theta)} \\ &= \lim_{\theta \rightarrow 0} \frac{-\tan(\pi \sin^2 \theta)}{\sin(2\pi \sin^2 \theta)} \\ &= \lim_{\theta \rightarrow 0} - \left(\frac{\tan(\pi \sin^2 \theta)}{\pi \sin^2 \theta} \right) \left(\frac{2\pi \sin^2 \theta}{\sin(2\pi \sin^2 \theta)} \right) \times \frac{1}{2} \\ &= \frac{-1}{2} \end{aligned}$$

Q5 (4)

$$\lim_{x \rightarrow 0} \frac{\left(x + \frac{x^3}{3!} \dots\right) - \left(x - \frac{x^3}{3!} \dots\right)}{3x^3} = \frac{1}{6}$$

So $6L + 1 = 2$